1. In November 2022, the official United Nations global human population estimate reached which milestone?

(A) 6 billion
(B) 7 billion
(C) 8 billion
(D) 10 billion

2. The theoretical population growth model where population increases by a constant percentage per unit time without resource limits is:

(A) Logistic growth
(B) Exponential (J-shaped) growth
(C) Linear growth
(D) Negative growth

3. The Sigmoid (S-shaped) population growth model that incorporates environmental carrying capacity (K) is:

(A) Malthusian growth
(B) Verhulst-Pearl Logistic growth
(C) Hyperbolic growth
(D) Geometric expansion

4. The rule of thumb used to estimate the doubling time (T_d) of a population growing at an annual rate of r% is the:

(A) Rule of 70 (T_d approx 70 / r)
(B) Rule of 10
(C) Rule of 100
(D) Rule of 3.14

5. If a nation's human population is growing at a constant annual rate of 2.0%, its doubling time is approximately:

(A) 14 years
(B) 35 years
(C) 70 years
(D) 140 years

6. Demographic Transition Theory describes the historical shift in human societies from:

(A) High birth and death rates to low birth and death rates during modernization
(B) Low birth rates to explosive birth rates
(C) Urbanization to agrarian living
(D) Nomadic hunting to feudalism

7. In the Demographic Transition Model, the rapid population explosion in Stage 2 is driven primarily by:

(A) Soaring birth rates while death rates remain constant
(B) A sharp decline in death rates (due to sanitation/medicine) while birth rates remain high
(C) Massive international immigration
(D) A sudden drop in life expectancy

8. The demographic condition where a nation experiences a high proportion of working-age adults relative to young and elderly dependents is:

(A) Demographic trap
(B) Demographic dividend
(C) Demographic winter
(D) Population crash

9. An expansive, broad-based triangular population age pyramid (typical of developing nations) indicates a population that is:

(A) Rapidly growing with a large proportion of young children
(B) Rapidly declining
(C) Completely stable with zero growth
(D) Heavily skewed toward elderly retirees

10. A constrictive population pyramid with a narrow base and bulging middle-aged tiers indicates:

(A) Rapid population explosion
(B) Declining birth rates and an aging population
(C) Equal birth and death rates
(D) Infinite carrying capacity

11. Which of the following is a classic water-borne bacterial infectious disease caused by Vibrio cholerae?

(A) Tuberculosis
(B) Cholera
(C) Malaria
(D) Rabies

12. Typhoid fever is a life-threatening water- and food-borne systemic infection caused by:

(A) Salmonella enterica serovar Typhi
(B) Escherichia coli
(C) Streptococcus pneumoniae
(D) Clostridium tetani

13. Malaria is a vector-borne protozoan disease transmitted to humans by the bite of infected female:

(A) Aedes aegypti mosquitoes
(B) Anopheles mosquitoes
(C) Culex pipiens mosquitoes
(D) Tsetse flies

14. Dengue fever, Chikungunya, and Zika virus infections are primarily transmitted by which mosquito vector?

(A) Anopheles stephensi
(B) Aedes aegypti (and Aedes albopictus)
(C) Culex quinquefasciatus
(D) Mansonia

15. Lymphatic filariasis (elephantiasis) is transmitted in tropical urban regions primarily by which mosquito vector?

(A) Culex mosquitoes
(B) Anopheles mosquitoes
(C) Aedes mosquitoes
(D) Sandflies

16. Synthetic chemical substances that interfere with the synthesis, secretion, or action of natural human hormones are:

(A) Mutagens
(B) Endocrine Disrupting Chemicals (EDCs)
(C) Neurotoxins
(D) Carcinogens alone

17. Bisphenol A (BPA), an endocrine disruptor widely used in polycarbonate plastics and epoxy can linings, mimics which natural hormone?

(A) Insulin
(B) Estrogen (17-beta-estradiol)
(C) Thyroxine
(D) Adrenaline

18. Phthalate plasticizers added to polyvinyl chloride (PVC) plastics act as endocrine disruptors that primarily disrupt:

(A) Thyroid hormone function
(B) Male androgen signaling and testicular development
(C) Glucagon release
(D) Vitamin D synthesis

19. The debilitating occupational lung disease caused by long-term inhalation of crystalline silica dust in stone quarries and mines is:

(A) Asbestosis
(B) Silicosis
(C) Byssinosis
(D) Bagassosis

20. Asbestosis is an occupational pulmonary fibrosis caused by inhaling asbestos mineral fibers, which frequently leads to fatal:

(A) Skin melanoma
(B) Malignant mesothelioma of the pleura
(C) Bone osteosarcoma
(D) Pancreatic cancer

21. Black Lung Disease (Coal Workers' Pneumoconiosis, CWP) is an occupational chronic lung disease caused by inhaling:

(A) Cotton dust
(B) Coal mine carbon dust
(C) Bagasse sugar cane fibers
(D) Wood saw dust

22. Byssinosis is an occupational respiratory disease occurring in textile workers exposed chronically to:

(A) Raw cotton, flax, and hemp dust
(B) Silica quartz sand
(C) Asbestos fibers
(D) Lead fumes

23. Infectious diseases that naturally transmit from vertebrate animals to humans (e.g., Rabies, Nipah, COVID-19) are termed:

(A) Nosocomial infections
(B) Zoonoses (Zoonotic diseases)
(C) Iatrogenic diseases
(D) Autoimmune disorders

24. The collaborative, multi-sectoral global approach recognizing that human health is interconnected with animal health and the environment is:

(A) The One Health Approach
(B) The Anthropocentric Initiative
(C) The Pure Clinic Model
(D) The Biochemical Protocol

25. The historic Chipko Movement for forest conservation was initiated in 1973 in which Himalayan region (now Uttarakhand)?

(A) Chamoli District (Reni Village)
(B) Shimla
(C) Darjeeling
(D) Srinagar

26. Who among the following was a renowned leader of the Chipko Movement who popularized the slogan 'Ecology is permanent economy'?

(A) Medha Patkar
(B) Sunderlal Bahuguna
(C) Baba Amte
(D) Pandurang Hegde

27. Which brave rural woman led the village women of Reni in March 1974 to hug the trees, forcing commercial loggers to retreat?

(A) Gaura Devi
(B) Amrita Devi
(C) Medha Patkar
(D) Vandana Shiva

28. The Appiko Movement, launched in September 1983 as the southern counterpart to Chipko, originated in which Indian state?

(A) Kerala
(B) Karnataka (Uttara Kannada district)
(C) Tamil Nadu
(D) Andhra Pradesh

29. Who was the primary grassroots organizer who led the Appiko Movement in the Western Ghats of Karnataka?

(A) Sunderlal Bahuguna
(B) Pandurang Hegde
(C) Chandi Prasad Bhatt
(D) Medha Patkar

30. The historic Silent Valley Movement in Kerala (1970s) was organized to protect a pristine tropical rainforest from a proposed:

(A) Open-pit bauxite mine
(B) Hydroelectric dam project across the Kunthipuzha River
(C) Nuclear power plant
(D) Paper pulp factory

31. Which prominent grassroots voluntary mass science movement in Kerala led the public campaign to save Silent Valley?

(A) Kerala Sasthra Sahithya Parishad (KSSP)
(B) Bishnoi Sabha
(C) Chipko Kendra
(D) Narmada Sangharsh Samiti

32. The Narmada Bachao Andolan (NBA), founded in 1985, mobilized public protests against massive tribal displacement caused by the:

(A) Tehri Dam
(B) Sardar Sarovar Dam
(C) Bhakra Nangal Dam
(D) Hirakud Dam

33. Who is the prominent social activist and environmental leader who has spearheaded the Narmada Bachao Andolan since its inception?

(A) Gaura Devi
(B) Medha Patkar
(C) Amrita Devi
(D) Arundhati Roy alone

34. The historic martyrdom of Amrita Devi and 363 Bishnoi community members sacrificing their lives to protect Khejri trees occurred in 1730 at:

(A) Ranthambore
(B) Khejarli Village (near Jodhpur, Rajasthan)
(C) Bikaner
(D) Udaipur

35. The philosophical viewpoint that views humans as the separate, central, and most important entities in the universe is:

(A) Biocentrism
(B) Anthropocentrism
(C) Ecocentrism
(D) Deep ecology

36. The ethical perspective that asserts that all individual living organisms have intrinsic moral worth regardless of utility to humans is:

(A) Anthropocentrism
(B) Biocentrism
(C) Social Darwinism
(D) Utilitarianism

37. The environmental philosophy that grants moral standing to entire ecological systems, species, and abiotic components is:

(A) Ecocentrism
(B) Anthropocentrism
(C) Egoism
(D) Hedonism

38. The philosophical concept of 'Deep Ecology' was formulated in 1973 by Norwegian philosopher:

(A) Arne Naess
(B) Aldo Leopold
(C) Garrett Hardin
(D) John Muir

39. Aldo Leopold's famous 'Land Ethic' asserts: 'A thing is right when it tends to preserve the integrity, stability, and beauty of the:'

(A) Human economic market
(B) Biotic community. It is wrong when it tends otherwise.
(C) Mining corporation
(D) Agricultural grain yield

40. The environmental justice concept that requires fair treatment and equal distribution of environmental resources within the present living generation is:

(A) Inter-generational equity
(B) Intra-generational equity
(C) Future discounting
(D) Eminent domain

41. The Verhulst-Pearl Logistic population growth equation is formulated mathematically as:

(A) $dN/dt = r * N * [(K - N) / K]$
(B) $dN/dt = r * N$
(C) $dN/dt = K * N / r$
(D) $dN/dt = r * (K - N)$

42. In the Verhulst-Pearl Logistic equation, the rate of population growth (dN/dt) reaches its absolute maximum when population size N equals:

(A) Zero
(B) K / 2 (half of the carrying capacity)
(C) K (at the carrying capacity asymptote)
(D) 2 * K

43. The biological population size (N = K / 2) at which population growth rate is maximized forms the theoretical basis in resource management for:

(A) Minimum Viable Population
(B) Maximum Sustainable Yield (MSY)
(C) Total Allowable Extinction
(D) Zero Population Growth

44. In environmental health risk assessment, the Chronic Daily Intake (CDI, in mg/kg-day) is calculated by which formula?

(A) (C * IR * EF * ED) / (BW * AT)
(B) (C * BW) / (IR * ED)
(C) (IR * EF) / (C * BW * AT)
(D) (C * ED) / (IR * BW)

45. In the CDI risk equation, what do the parameters 'IR', 'BW', and 'AT' represent?

(A) Infrared, Boiling Water, Air Temperature
(B) Ingestion/Inhalation Rate, Body Weight, and Averaging Time
(C) Initial Rate, Biomass Weight, and Atmospheric Transfer
(D) Internal Resistance, Body Width, and Total Area

46. The non-carcinogenic toxic risk of a chemical pollutant is quantified by the Hazard Quotient (HQ), defined as:

(A) HQ = Chronic Daily Intake (CDI) / Reference Dose (RfD)
(B) HQ = Reference Dose / CDI
(C) HQ = CDI * Cancer Slope Factor
(D) HQ = Body Weight / Exposure Duration

47. In toxicological risk assessment, a Hazard Quotient HQ > 1.0 indicates that the exposed human population:

(A) Faces zero toxicological risk
(B) Faces potential adverse, non-carcinogenic health effects exceeding safe exposure thresholds
(C) Is 100% immune to toxins
(D) Will undergo immediate death within seconds

48. The incremental Lifetime Cancer Risk (LCR) for exposure to a carcinogenic chemical pollutant is calculated by:

(A) LCR = CDI * Slope Factor (SF)
(B) LCR = CDI / Slope Factor
(C) LCR = RfD * CDI
(D) LCR = Hazard Quotient / 100

49. In global health epidemiology, the metric 'Disability-Adjusted Life Year' (DALY) represents the sum of:

(A) Years of Life Lost (YLL) due to premature mortality + Years Lived with Disability (YLD)
(B) Total hospital days + Total medicine cost
(C) Total deaths multiplied by average age
(D) Gross Domestic Product loss per patient

50. Inter-generational equity in environmental jurisprudence mandates that the present human generation holds the Earth in trust for:

(A) Future human generations, ensuring the planet is passed on in no worse condition than received
(B) Present corporate shareholders exclusively
(C) Immediate municipal consumption
(D) Current military operations

51. In Stage 1 of the Demographic Transition Model (Pre-Industrial Society), population dynamics are characterized by:

(A) High crude birth rates and high crude death rates, yielding a low, fluctuating population growth
(B) Rapidly collapsing birth rates
(C) Zero infant mortality
(D) Negative population growth

52. Stage 4 of the Demographic Transition Model (Modern Industrial Society) is defined by:

(A) Low birth rates and low death rates, resulting in a stable, stationary population
(B) Explosive population surge
(C) Famine and epidemic mortality
(D) Doubling of fertility rates

53. Stage 5 of the Demographic Transition Model describes modern post-industrial societies where:

(A) Crude birth rates fall below crude death rates, leading to natural population decline (e.g., Japan, Italy)
(B) Birth rates surge to 50 per 1,000
(C) Death rates quadruple
(D) Carrying capacity becomes infinite

54. The average number of children that would be born to a woman over her lifetime if she experiences current age-specific fertility rates is:

(A) Crude Birth Rate
(B) Total Fertility Rate (TFR)
(C) General Fertility Ratio
(D) Net Reproduction Rate

55. In demography, the Replacement-Level Fertility Rate in developed nations with low infant mortality is standardized at approximately:

(A) 1.5 children per woman
(B) 2.1 children per woman
(C) 3.0 children per woman
(D) 4.2 children per woman

56. According to India's National Family Health Survey-5 (NFHS-5, 2019-21), India's national Total Fertility Rate (TFR) has declined to:

(A) 2.0 children per woman (below the replacement level of 2.1)
(B) 2.7 children per woman
(C) 3.4 children per woman
(D) 1.2 children per woman

57. The Natural Increase Rate (NIR) of a human population is calculated directly from Crude Birth Rate (CBR) and Crude Death Rate (CDR) by:

(A) NIR (%) = (CBR - CDR) / 10
(B) NIR (%) = CBR * CDR
(C) NIR (%) = (CBR + CDR) / 100
(D) NIR (%) = CDR / CBR

58. The One Health quadripartite global alliance includes WHO, WOAH (Animal Health), FAO, and which environmental agency?

(A) UNEP (United Nations Environment Programme)
(B) WTO
(C) IMF
(D) UNESCO alone

59. Zoonotic viral disease spillover from wild bats to humans (e.g., Nipah virus outbreaks in Kerala) is primarily accelerated by:

(A) Deforestation, fruit orchard fragmentation, and habitat destruction forcing bats into agricultural zones
(B) Industrial robotics
(C) Oceanic thermal upwelling
(D) Clean solar energy installations

60. Endocrine disrupting chemicals (EDCs) like Bisphenol A and phthalates exert toxic effects at ultra-low concentrations by:

(A) Binding directly to nuclear hormone receptors (ER, AR), triggering abnormal gene transcription
(B) Precipitating all body water
(C) Oxidizing bone calcium into lime
(D) Freezing cellular cytoplasm

61. The Environmental Information System (ENVIS) was established in India in 1982 by the Central Government to serve as a decentralized network for:

(A) Storing, retrieving, and disseminating environmental information to policymakers, researchers, and the public
(B) Issuing industrial criminal warrants
(C) Conducting secret nuclear research
(D) Levying national sales taxes

62. In satellite remote sensing of vegetation, the Normalized Difference Vegetation Index (NDVI) is mathematically calculated as:

(A) NDVI = (NIR - Red) / (NIR + Red)
(B) NDVI = (NIR + Red) / (NIR - Red)
(C) NDVI = NIR * Red
(D) NDVI = (Green - Blue) / (Green + Blue)

63. Healthy, dense photosynthetic vegetation exhibits high NDVI values (typically 0.6 to 0.85) because chlorophyll:

(A) Absorbs red visible light for photosynthesis and strongly reflects near-infrared (NIR) radiation
(B) Reflects red light and absorbs all NIR
(C) Absorbs all wavelengths equally
(D) Emits X-rays

64. Under the Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 (LARR Act):

(A) Prior Social Impact Assessment (SIA) is mandatory before acquiring land
(B) Land can be seized without financial compensation
(C) Local Gram Sabhas have zero right to consultation
(D) Private corporations can forcibly evict residents without notice

65. Under the LARR Act 2013, the financial compensation mandated for acquired land in rural agricultural areas is:

(A) Up to four times the market value of the land
(B) Half of the assessed market value
(C) Exactly 100 rupees per acre
(D) Zero compensation if required for public roads

66. The Eight-Point Platform of Deep Ecology formulated by Arne Naess and George Sessions explicitly asserts that:

(A) Human beings have no right to reduce the richness and diversity of life except to satisfy vital needs
(B) Nature exists exclusively for unlimited human industrial consumption
(C) All animal species should be domesticated into livestock
(D) Technological engineering will replace all wild forests

67. The philosophical school of 'Ecofeminism' popularized by Vandana Shiva and Maria Mies draws structural connections between:

(A) The patriarchal subjugation of women and the capitalist exploitation and degradation of nature
(B) Urban traffic congestion and solar flares
(C) Desertification and volcanic lava
(D) Deep sea fishing and space satellites

68. In 1991, the Supreme Court of India in the landmark M.C. Mehta v. Union of India case issued a historic directive mandating:

(A) Environmental education as a compulsory subject across all schools, colleges, and university degrees in India
(B) A complete shutdown of all Indian universities
(C) The nationalization of all private farmland
(D) The banning of all automobiles nationwide

69. According to the World Health Organization (WHO), there is NO identified safe threshold for lead (Pb) exposure in children, with BLL > 3.5 ug/dL causing:

(A) Permanent cognitive deficits, reduced intelligence quotient (IQ), and behavioral attention disorders
(B) Temporary freckles
(C) Improved hearing acuity
(D) Rapid muscle hypertrophy

70. In the historic 2013 Supreme Court ruling on the Niyamgiri hills in Odisha, the Court held that the Dongria Kondh tribal Gram Sabhas had the legal right under the Forest Rights Act to:

(A) Decide whether bauxite mining by Vedanta on their sacred mountain (Niyamraja) would violate their religious and cultural rights
(B) Sell the sacred mountain to private foreign hedge funds
(C) Declare independence from the Indian Union
(D) Construct international airports

71. The Bishnoi religious conservation ethic was founded in the 15th century by Guru Jambheshwar, who codified how many sacred rules?

(A) 10 Commandments
(B) 29 Rules (Bish = 20, Noi = 9)
(C) 50 Principles
(D) 100 Vows

72. The Khejri tree (Prosopis cineraria), protected with their lives by the Bishnoi community, is ecologically vital in arid Rajasthan because it:

(A) Fixes nitrogen, stabilizes sand dunes, and provides nutritious green fodder during extreme drought
(B) Produces toxic sap that kills all insects
(C) Absorbs all groundwater within 50 miles
(D) Burns spontaneously in daylight

73. In global citizen science, online platforms like eBird and iNaturalist advance biodiversity conservation by:

(A) Aggregating millions of geotagged biological observations from amateur naturalists into open-access scientific databases
(B) Selling rare birds to private collectors
(C) Issuing hunting permits online
(D) Tracking military aircraft movements

74. In occupational health, prolonged inhalation of crystalline silica dust induces progressive silicosis because silica particles inside alveolar macrophages:

(A) Rupture phagolysosomes, triggering chronic release of fibrogenic cytokines (TNF-alpha, IL-1) and collagen scarring
(B) Dissolve harmlessly into blood sugar
(C) Transform macrophages into red blood cells
(D) Evaporate into helium gas

75. Arne Naess contrasted 'Shallow Ecology' with 'Deep Ecology', defining Shallow Ecology as a movement that:

(A) Fights pollution and resource depletion merely to protect the health and affluence of humans in developed nations
(B) Advocates the extinction of the human race
(C) Rejects all modern medicine
(D) Prohibits all agriculture

76. A developing district records a Crude Birth Rate (CBR) of 32 per 1,000 and a Crude Death Rate (CDR) of 8 per 1,000. Using the Rule of 70 [T_d = 70 / NIR%], calculate the doubling time of the population:

(A) 14.6 years
(B) 29.2 years
(C) 35.0 years
(D) 58.3 years

77. A biological population in a managed habitat follows logistic growth with carrying capacity K = 2,000 individuals and intrinsic growth rate r = 0.10 yr^-1. Calculate the instantaneous population growth rate dN/dt when N = 500 individuals [dN/dt = r*N*(1 - N/K)]:

(A) 25 individuals/year
(B) 37.5 individuals/year
(C) 50.0 individuals/year
(D) 75.0 individuals/year

78. For the population in Question 77, what is the Maximum Sustainable Yield (MSY) harvesting rate achieved when the population is maintained at N = K / 2 = 1,000 individuals?

(A) 25.0 individuals/year
(B) 37.5 individuals/year
(C) 50.0 individuals/year
(D) 100.0 individuals/year

79. An adult weighing BW = 70 kg drinks IR = 2.0 L/day of groundwater containing C = 0.05 mg/L of dissolved Arsenic over an exposure duration ED = 30 years (EF = 350 days/yr, AT = 30 * 365 = 10,950 days). Calculate the Chronic Daily Intake (CDI = [C * IR * EF * ED] / [BW * AT]):

(A) 0.00055 mg/kg-day
(B) 0.00137 mg/kg-day
(C) 0.00500 mg/kg-day
(D) 0.01250 mg/kg-day

80. For the individual in Question 79, the oral Reference Dose (RfD) for arsenic is 0.00030 mg/kg-day. What is the non-carcinogenic Hazard Quotient (HQ = CDI / RfD)?

(A) 0.46
(B) 1.85
(C) 4.57 (Unacceptable toxic risk)
(D) 10.50

81. For the individual in Question 79, the lifetime average daily dose for carcinogenic risk (averaging time AT = 70 years * 365 days = 25,550 days) is CDI_carc = 0.000587 mg/kg-day. If the oral Cancer Slope Factor (SF) for arsenic is 1.5 (mg/kg-day)^-1, what is the incremental Lifetime Cancer Risk (LCR = CDI * SF)?

(A) 1.5 * 10^-6
(B) 8.8 * 10^-4 (approx 1 in 1,136)
(C) 5.0 * 10^-2
(D) 1.0 * 10^-1

82. In an endemic fluorosis village of 1,000 residents, 50 individuals suffer from severe skeletal fluorosis (Disability Weight DW = 0.35) for an average duration of 20 years (YLD = 50 * 0.35 * 20 = 350). Additionally, 5 individuals die prematurely at age 45 (standard life expectancy = 75, YLL = 5 * 30 = 150). What is the total burden of disease in Disability-Adjusted Life Years (DALY = YLL + YLD)?

(A) 150 DALYs
(B) 350 DALYs
(C) 500 DALYs
(D) 1,000 DALYs

83. A satellite multispectral sensor records surface reflectance over an agricultural watershed: Near-Infrared band (NIR) reflectance = 0.60, and Red band reflectance = 0.10. Calculate the Normalized Difference Vegetation Index [NDVI = (NIR - Red) / (NIR + Red)]:

(A) 0.500
(B) 0.625
(C) 0.714
(D) 0.850

84. For a clear, deep freshwater lake, the satellite sensor records NIR reflectance = 0.02 and Red reflectance = 0.08. What is the calculated NDVI?

(A) -0.600 (Negative value indicating water body)
(B) +0.100
(C) +0.600
(D) 0.000

85. In a nation with constant population size N = 100 million and zero net migration, the Total Fertility Rate (TFR) stabilizes at 1.4 children per woman (below replacement level 2.1). In the long term, by what percentage does each successive generation contract [1 - (1.4 / 2.1)]?

(A) 10.0%
(B) 20.5%
(C) 33.3%
(D) 50.0%

86. A city has a total population of 1,000,000. Working-age adults (15 to 64 years) number 650,000. Children (0 to 14 years) number 250,000, and elderly retirees (65+ years) number 100,000. Calculate the Total Age Dependency Ratio [TDR = (Dependents / Working-Age) * 100]%:

(A) 35.0%
(B) 53.8%
(C) 65.0%
(D) 75.0%

87. Under the LARR Act 2013, the base market value of 10 hectares of acquired rural farmland is assessed at 20 lakh rupees per hectare (total base = 200 lakhs). With a rural market multiplication factor of 2.0 and mandatory 100% solatium, what is the total statutory compensation payable?

(A) 200 lakh rupees
(B) 400 lakh rupees
(C) 800 lakh rupees (4x base market value)
(D) 1,200 lakh rupees

88. A municipality with an initial population of 500,000 records 10,000 live births (CBR = 20/1000) and 4,000 deaths (CDR = 8/1000) in one year. Net annual immigration is 4,000 persons. What is the total net annual population growth rate (%)?

(A) 0.8%
(B) 1.2%
(C) 2.0%
(D) 3.5%

89. Thomas Malthus posited in 1798 that while agricultural food production increases at an arithmetic progression (1, 2, 3, 4, 5...), human population when unchecked expands at a:

(A) Logarithmic rate
(B) Geometric (exponential) progression (1, 2, 4, 8, 16...)
(C) Linear constant rate
(D) Negative regression

90. In environmental health epidemiology, Odds Ratio (OR) in a case-control study comparing silicosis cases to controls exposed to stone quarry dust is calculated from the contingency table as:

(A) OR = (a * d) / (b * c)
(B) OR = (a + b) / (c + d)
(C) OR = (a * b) / (c * d)
(D) OR = a / (a + b)

91. The blood lead reference value (BLRV) established by the US CDC for initiating clinical and public health interventions in young children is:

(A) 3.5 micrograms/dL
(B) 25.0 micrograms/dL
(C) 50.0 micrograms/dL
(D) 100.0 micrograms/dL

92. In the Verhulst-Pearl logistic growth model dN/dt = r*N*(1 - N/K), the term (1 - N/K) is biologically interpreted as the:

(A) Unutilized intrinsic growth potential
(B) Fraction of environmental carrying capacity remaining available for population expansion
(C) Total mortality rate of the species
(D) Fecundity rate per female

93. According to the M.C. Mehta Oleum Gas Leak ruling (1987), an enterprise engaged in hazardous industrial activity cannot escape liability using which common law defense?

(A) Act of God (Vis Major) or sabotage by an unauthorized third party
(B) Proof of valid government license
(C) Payment of local municipal taxes
(D) Absence of criminal intent

94. Under the Indian Forest Rights Act 2006, what is the maximum area of forest land that can be recognized as an Individual Forest Right title per nuclear family?

(A) 1.0 hectare
(B) 2.5 hectares
(C) 4.0 hectares
(D) 10.0 hectares

95. In air pollution dispersion modeling, the Ground Level Concentration C(x,y,0) directly downwind (y = 0) from an elevated chimney stack of effective height H is formulated by the Gaussian plume model as:

(A) C = [Q / (pi * u * sigma_y * sigma_z)] * exp[-0.5 * (H / sigma_z)^2]
(B) C = Q / (u * H)
(C) C = Q * u * sigma_y
(D) C = exp(-H / u)

96. In the Gaussian plume dispersion equation, what do the dispersion coefficients sigma_y and sigma_z represent?

(A) Gas density and temperature
(B) Standard deviations of horizontal and vertical plume concentration distributions (increasing with downwind distance x)
(C) Stack diameter and gas exit velocity
(D) Ambient air pressure and wind speed

97. A rural health survey finds that children consuming water with 4.0 mg/L fluoride experience a 60% prevalence of dental fluorosis compared to 5% in a low-fluoride control village (0.5 mg/L). What is the Relative Risk (RR) of dental fluorosis?

(A) 3.0
(B) 6.0
(C) 12.0
(D) 24.0

98. In environmental economics, the Hotelling's Rule for non-renewable mineral resources states that the net price (resource rent) of an exhaustible resource must increase over time at a rate equal to the:

(A) Rate of inflation
(B) Prevailing market rate of interest
(C) Rate of population growth
(D) Depletion rate of the mine

99. The 'Ecological Footprint of Nations' reveals that if all 8 billion humans on Earth lived with the per capita resource consumption of an average citizen of the United States, humanity would require approximately:

(A) 1.0 Earth
(B) 2.5 Earths
(C) 5.1 Earths
(D) 15.0 Earths

100. The landmark Brundtland Report concluded that poverty and environmental degradation are locked in a vicious self-reinforcing cycle, stating that:

(A) 'Poverty is both a major cause and effect of global environmental problems; therefore, environmental protection cannot be divorced from economic development.'
(B) 'Economic development must be permanently halted to save the planet.'
(C) 'Only wealthy nations have the right to consume fossil fuels.'
(D) 'Industrial pollution is completely harmless to poor communities.'

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Formulate the Malthusian Population Theory and derive the Verhulst-Pearl Logistic Growth Model, explaining the inflection point and Maximum Sustainable Yield.

Q2. Analyze the five stages of the Demographic Transition Model, detailing birth rate, death rate, population growth, and the Demographic Dividend.

Q3. Examine the geometric morphology of Population Age Pyramids, contrasting Expansive, Constrictive, and Stationary profiles and their societal impacts.

Q4. Formulate the Epidemiological Triad (Agent, Host, Environment), and explain the Environmental Determinants of human health and health equity.

Q5. Detail the etiology, molecular pathophysiology, and transmission mechanics of Water-Borne Bacterial Diseases: Cholera and Typhoid Fever.

Q6. Analyze the transmission cycles and climate sensitivity of Vector-Borne Diseases in tropical agroecosystems: Dengue and Malaria.

Q7. Explain the molecular toxicology of Endocrine Disrupting Chemicals (EDCs), detailing the receptor-mediated actions of Bisphenol A (BPA) and Phthalates.

Q8. Detail the pathophysiology of Occupational Pulmonary Fibrosis, analyzing alveolar macrophage activation in Silicosis and Asbestosis.

Q9. Analyze the drivers of Zoonotic Disease Emergence and viral spillover, and formulate the collaborative architecture of the One Health approach.

Q10. Examine the history, grassroots leadership, and policy impacts of the Chipko Movement (1973), evaluating its eco-feminist dimensions.

Q11. Detail the Appiko Movement (1983) in the Western Ghats of Karnataka, analyzing its operational methodology and forest conservation outcomes.

Q12. Analyze the Silent Valley Movement (1970s) in Kerala, detailing the ecological activism of KSSP and the preservation of tropical rainforests.

Q13. Examine the Narmada Bachao Andolan (NBA, 1985), focusing on big dam controversies, Sardar Sarovar submergence, and tribal human rights.

Q14. Detail the environmental ethos of the Bishnoi Community, the 29 Principles of Guru Jambheshwar, and the 1730 Khejarli martyrdom.

Q15. Compare the philosophical paradigms of Anthropocentrism, Biocentrism, and Ecocentrism, evaluating Arne Naess' Deep Ecology framework.

Q16. Formulate the four-step Environmental Health Risk Assessment (EHRA) framework defined by the US National Research Council (NRC).

Q17. Explain the derivation of the Reference Dose (RfD) for non-carcinogens, defining NOAEL, LOAEL, and Uncertainty (Safety) Factors.

Q18. Analyze the mathematical metrics used in environmental health epidemiology: Relative Risk (RR), Odds Ratio (OR), and Disability-Adjusted Life Years (DALY).

Q19. Detail the molecular toxicology of Lead (Pb) poisoning, explaining ALAD enzyme inhibition, calcium mimicry, and cognitive neurodevelopmental deficits.

Q20. Examine the low-dose non-monotonic dynamics of Bisphenol A (BPA) and evaluate evidence for Epigenetic Transgenerational Inheritance.

Q21. Analyze the principles of Environmental Justice, distinguishing Inter-Generational and Intra-Generational Equity, and the Basel Convention.

Q22. Detail the architecture and operational mandate of the Environmental Information System (ENVIS) in India.

Q23. Explain the applications of Satellite Remote Sensing in environmental management, detailing multispectral sensors, NDVI, and forest monitoring.

Q24. Analyze the role of Geographic Information Systems (GIS) in environmental decision-making, spatial overlay analysis, and watershed modeling.

Q25. Examine the role of Environmental Education in India, analyzing the Supreme Court directives in the M.C. Mehta litigation and Eco-Club programs.

Q26. Analyze the role of Non-Governmental Organizations (NGOs) and Public Interest Litigation (PIL) in Indian environmental governance, detailing landmark cases.

Q27. Examine the socio-economic and cultural impacts of Development-Induced Displacement, evaluating the marginalization of Project-Affected Persons (PAPs).

Q28. Detail the statutory framework of the Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 (LARR Act).

Q29. Formulate the Gaussian Plume Model for atmospheric air pollution dispersion, defining Pasquill-Gifford stability classes and inversion effects.

Q30. Detail the engineering principles of Circular Sustainable Cities, evaluating the Kalundborg Industrial Symbiosis paradigm and Urban Metabolism.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[C]	**Q21:**	[B]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[B]
Q2:	[B]	**Q22:**	[A]	**Q42:**	[B]	**Q62:**	[A]	**Q82:**	[C]
Q3:	[B]	**Q23:**	[B]	**Q43:**	[B]	**Q63:**	[A]	**Q83:**	[C]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[B]	**Q25:**	[A]	**Q45:**	[B]	**Q65:**	[A]	**Q85:**	[C]
Q6:	[A]	**Q26:**	[B]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[B]
Q7:	[B]	**Q27:**	[A]	**Q47:**	[B]	**Q67:**	[A]	**Q87:**	[C]
Q8:	[B]	**Q28:**	[B]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[C]
Q9:	[A]	**Q29:**	[B]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[B]
Q10:	[B]	**Q30:**	[B]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[B]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[B]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[B]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[B]
Q13:	[B]	**Q33:**	[B]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[B]	**Q34:**	[B]	**Q54:**	[B]	**Q74:**	[A]	**Q94:**	[C]
Q15:	[A]	**Q35:**	[B]	**Q55:**	[B]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[B]	**Q36:**	[B]	**Q56:**	[A]	**Q76:**	[B]	**Q96:**	[B]
Q17:	[B]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[B]	**Q97:**	[C]
Q18:	[B]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[C]	**Q98:**	[B]
Q19:	[B]	**Q39:**	[B]	**Q59:**	[A]	**Q79:**	[B]	**Q99:**	[C]
Q20:	[B]	**Q40:**	[B]	**Q60:**	[A]	**Q80:**	[C]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

- Stage 1: High Stationary (Pre-Industrial Society):
* Crude Birth Rate (CBR): Very high (35 to 45 per 1,000), driven by high infant mortality, cultural norms, and agrarian labor needs.
* Crude Death Rate (CDR): Very high and erratic (30 to 45 per 1,000), driven by recurrent famines, infectious epidemics (cholera, plague), and primitive sanitation.
* Population Growth: Very low, near-zero long-term growth with wild cyclical fluctuations.
- Stage 2: Early Expanding (Developing Society):
* CDR collapses rapidly (falling to 15-20 per 1,000) due to clean drinking water, municipal sewage systems, antibiotics, vaccination, and food transport.
* CBR remains stubbornly high due to cultural inertia and lack of family planning.
* Population Dynamics: Rapid, explosive population growth! (The historical phase responsible for global population surges).
- Stage 3: Late Expanding (Maturing Developing Society):
* CBR begins to decline significantly (dropping to 15-20 per 1,000) driven by urbanization, female literacy, entry of women into the formal workforce, increased cost of child-rearing, and widespread contraception.
* CDR continues to decline slowly (stabilizing at 8-10 per 1,000).
* Population Growth: Continues to grow, but the rate of expansion decelerates.
- Stage 4: Low Stationary (Developed Industrial Society):
* Both CBR and CDR stabilize at low levels (10 to 12 per 1,000).
* Total Fertility Rate (TFR) hovers near replacement level (~2.1). Population stabilizes into a stationary equilibrium.
- Stage 5: Natural Decline (Post-Industrial Society, e.g., Japan, Italy, Germany):
* CBR falls well below CDR (TFR drops to 1.2 to 1.5). Death rates rise slightly due to an aging demographic structure.
* Population experiences persistent, natural contraction.
3. The Demographic Dividend:
- A strategic economic window occurring during Stages 2 and 3 when the birth rate decline produces a bulge in the working-age population (15 to 64 years) while the Dependency Ratio (proportion of dependent children and elderly) falls to an all-time low.
- If accompanied by investments in education, healthcare, and employment creation, this demographic structure unleashes rapid GDP growth and national capital accumulation.

Solution to Question 3: >> Examine the geometric morphology of Population Age Pyramids, contrasting Expansive, Constrictive, and Stationary profiles and their s

(30-55 years), and broad toward older cohorts.
* Demographic Characteristics: Low and falling birth rates (TFR < 1.6), low mortality, high life expectancy. Proportion of children is smaller than the reproductive adult cohort.
* Typical Nations: Japan, Italy, South Korea, Germany.
* Socio-Economic Challenges: Rapid population aging (graying society), escalating geriatric healthcare and pension expenditures, severe industrial labor shortages, and shrinking consumer tax bases.
- 3. Stationary / Stable Population Pyramid (Rectangular / Bell Shape):
* Geometric Profile: Uniform, near-vertical sides with approximately equal widths across all age cohorts from infancy through middle age, tapering gently only at advanced ages (75+ years).
* Demographic Characteristics: Low birth and death rates with TFR hovering near replacement level (2.1). Zero or near-zero population growth.
* Typical Nations: Sweden, France, Norway, United States.
* Socio-Economic Stability: Balanced dependency ratios, predictable educational and labor requirements, and manageable social security burdens.

>>Solution to Question 4: Formulate the Epidemiological Triad (Agent, Host, Environment), and explain the Environmental Determinants of human health and hea

disproportionately in close proximity to polluting industrial plants, municipal landfills, and toxic effluent drains, suffering elevated morbidity while possessing minimal financial resources to access curative medical care.

>>Solution to Question 5: Detail the etiology, molecular pathophysiology, and transmission mechanics of Water-Borne Bacterial Diseases: Cholera and Typhoid F

1. Definition of Endocrine Disruptors (US EPA):
Exogenous chemical substances or mixtures that alter one or more functions of the endocrine system and consequently cause adverse health effects in an intact organism, its progeny, or populations.
- Non-Monotonic Dose-Response Curves (NMDRCs):
Unlike classical industrial poisons where toxicity follows Paracelsus' linear dogma ('the dose makes the poison'), EDCs frequently exhibit U-shaped or inverted U-shaped dose-response curves, exerting potent physiological and epigenetic disruptions at ultra-low nanomolar concentrations (parts-per-billion or parts-per-trillion) while displaying minimal toxicity at high doses!
2. Bisphenol A (BPA): Molecular Mimicry and Cellular Mechanisms:
- Chemical Identity: 2,2-bis(4-hydroxyphenyl)propane. Widely synthesized as a monomer in the production of polycarbonate plastics (baby bottles, food containers) and epoxy resin coatings lining metal food and beverage cans.
- Xeno-Estrogenic Activity:
* BPA contains two phenolic rings structurally analogous to the natural female steroid hormone 17-beta-estradiol (E2).
* It binds to nuclear Estrogen Receptors alpha and beta (ER-alpha and ER-beta) and membrane-bound G-protein coupled estrogen receptor 1 (GPER1), inducing allosteric dimerization and recruiting co-activator proteins that trigger premature estrogen-responsive gene transcription.
* Developmental & Reproductive Pathologies: Early fetal exposure induces altered prostate gland morphogenesis, premature puberty in females, reduced sperm counts in males, polycystic ovarian syndrome (PCOS), and elevated lifetime risk of hormone-dependent breast and prostate cancers.
3. Phthalate Esters: Anti-Androgenic and Metabolic Disruption:
- Chemical Identity: Dialkyl or alkyl aryl esters of 1,2-benzenedicarboxylic acid [e.g., Di(2-ethylhexyl) phthalate, DEHP; Di-n-butyl phthalate, DBP]. Used as plasticizers to confer flexibility to rigid polyvinyl chloride (PVC) plastics.
- Anti-Androgenic Endocrine Mechanism:
* Phthalates are not chemically bound to the PVC polymer matrix; they leach continuously into food, milk, and medical IV tubing.
* In fetal male Leydig cells during the critical developmental 'masculinization programming window,' phthalate metabolites down-regulate the expression of the steroidogenic acute regulatory (StAR) protein and cytochrome P450 side-chain cleavage enzyme (CYP11A1), shutting down fetal testosterone biosynthesis.
* Phthalate Syndrome: Induces congenital urogenital abnormalities: Cryptorchidism (undescended testes), Hypospadias (abnormal urethral opening), shortened Anogenital Distance (AGD), and adult testicular dysgenesis syndrome.

>>Solution to Question 8: Detail the pathophysiology of Occupational Pulmonary Fibrosis, analyzing alveolar macrophage activation in Silicosis and Asbestosis.

in stone crushing, granite mining, foundry work, sandblasting, and ceramic manufacture.
- Macrophage Phagocytosis and Lysosomal Rupture:
* Alveolar macrophages engulf crystalline silica particles via scavenger receptors.
* The highly reactive, cleaved silanol (-Si-OH) surface groups on quartz crystals generate massive hydroxyl and superoxide free radicals, peroxidizing membrane lipids.
* Phagolysosomes destabilize and rupture, spilling acidic hydrolases and silica crystals directly into the macrophage cytosol.
- NLRP3 Inflammasome Activation:
* Cytosolic silica activates the NLRP3 inflammasome multiprotein complex, which cleaves pro-caspase-1 into active caspase-1.
* Caspase-1 enzymatically converts pro-interleukin-1-beta into active Interleukin-1-beta (IL-1-beta), releasing high concentrations of IL-1-beta and Tumor Necrosis Factor-alpha (TNF-alpha).
* The macrophage undergoes lytic cell death (pyroptosis), releasing the undigested silica particle back into the alveolar space to be engulfed by fresh macrophages, establishing a permanent, self-sustaining inflammatory cascade!
- Fibrogenesis and Silicotic Nodules:
* Persistent cytokine signaling recruits interstitial fibroblasts, which proliferate and deposit concentric whorls of dense, hyalinized type I collagen around central quartz cores (Silicotic Nodules).
* Silicotic nodules coalesce into Massive Progressive Fibrosis, destroying alveolar gas exchange, causing irreversible restrictive respiratory failure and elevating tuberculosis risk 30-fold.
3. Pathogenesis of Asbestosis and Mesothelioma:
- Asbestos consists of fibrous silicate minerals: Serpentines (curly chrysotile / white asbestos) and Amphiboles (needle-like rigid crocidolite / blue asbestos and amosite).
- Frustrated Phagocytosis: Amphibole fibers possess high aspect ratios (length:width > 3:1) and lengths exceeding 10 to 20 micrometers. Alveolar macrophages cannot fully engulf these long needles ('frustrated phagocytosis'), leaking chronic proteolytic enzymes, generating persistent pleural inflammation, pulmonary fibrosis, and inducing malignant DNA double-strand breaks that lead to fatal Pleural Mesothelioma.

>>**Solution to Question 9:** Analyze the drivers of Zoonotic Disease Emergence and viral spillover, and formulate the collaborative architecture of the One Health a

before jumping to human farmworkers (e.g., Nipah virus passing from fruit bats to swine to slaughterhouse workers in Malaysia).

- 3. Commercial Wildlife Trade and Wet Markets:

Crowding diverse wild species (bats, civets, pangolins) together under stressful, unhygienic conditions in commercial wet markets facilitates inter-species viral recombination and high-volume human aerosol exposure.

3. The 'One Health' Integrated Transdisciplinary Paradigm:

Endorsed by the Quadripartite Alliance (WHO, WOAH, FAO, and UNEP), One Health asserts that human medicine CANNOT function in isolation from veterinary medicine and environmental conservation:

- Core Pillars of Action:

* 1. Upstream Ecological Surveillance: Monitoring viral diversity (virome discovery) in natural wildlife reservoir populations before spillover occurs.
* 2. Integrated Biosecurity at the Animal-Human Interface: Enforcing strict biosecurity barriers on livestock farms, ending illegal bushmeat trafficking, and closing commercial wildlife wet markets.
* 3. Joint Multi-Sectoral Outbreak Response: Unifying public health departments, veterinary epidemiologists, and forestry wildlife rangers under centralized command structures to detect and contain novel pathogen outbreaks at the source.

>>**Solution to Question 10:** Examine the history, grassroots leadership, and policy impacts of the Chipko Movement (1973), evaluating its eco-feminist dimensions,

- Popularized the landmark eco-centric slogan:
'What do the forests bear? Soil, water, and pure air! Soil, water, and pure air are the basis of life!'
5. Policy and Legislative Triumphs:
- In 1980, Prime Minister Indira Gandhi met with Bahuguna and enacted a historic 15-year statutory ban on the felling of green trees above 1,000 meters elevation in the Himalayas.
- Directly catalyzed the passage of the landmark national Forest (Conservation) Act, 1980.

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possessing over 50 million years of unbroken evolutionary history. It harbors core breeding populations of the globally endangered Lion-tailed Macaque (Macaca silenus).

2. The Hydroelectric Threat:

In 1973, the Kerala State Electricity Board (KSEB) proposed the construction of a 131-meter-high concrete gravity dam across the pristine Kunthipuzha River, submerging 830 hectares of primary tropical rainforest to generate 120 MW of peak hydroelectric power.

3. Science Activism by Kerala Sasthra Sahithya Parishad (KSSP):

- KSSP (Kerala Science Literature Movement), a voluntary mass organization of school teachers, university scientists, and writers, transformed the campaign from an emotional protest into a rigorous, scientifically grounded environmental battle.
- Interdisciplinary Technoeconomic Survey:

KSSP scientists published an exhaustive alternative assessment proving that:

* The dam would destroy rare, irreplaceable endemic biodiversity and primitive floral germplasm.
* The actual electricity generated (120 MW) would contribute < 10% to Kerala's grid and could be achieved far more cheaply by reducing transmission and distribution (T&D) losses and upgrading existing thermal plants.
* The promised irrigation benefits were marginal compared to the permanent destruction of the microclimate.

4. Creative Public Communication Campaigns:

- KSSP mobilized mass public awareness through street plays, traveling song processions (jathas), newspaper columns, and environmental poetry (led by renowned poet Sugathakumari).
- Prominent national scientists (Dr. Salim Ali, Satish Dhawan, M.S. Swaminathan) submitted expert reports to the Central Government urging total preservation.

5. National Policy Triumph (1985):

- In 1983, Prime Minister Indira Gandhi formally declared that the Silent Valley hydroelectric project was cancelled to protect the ecosystem.
- In November 1985, Prime Minister Rajiv Gandhi officially inaugurated the Silent Valley National Park, and the valley subsequently became the core protected zone of the Nilgiri Biosphere Reserve.

>>Solution to Question 13: Examine the Narmada Bachao Andolan (NBA, 1985), focusing on big dam controversies, Sardar Sarovar submergence, and tribal hum

3. The World Bank Independent Morse Commission (1992):
- NBA organized massive non-violent marches, hunger strikes, and Jal Satyagraha (protesters refusing to leave their ancestral huts as floodwaters rose, standing up to their necks in rising dam waters).
- International Pressure forced the World Bank to appoint an independent evaluation mission headed by Bradford Morse.
- The Morse Report (1992) issued a scathing condemnation of the Sardar Sarovar Project, documenting systematic failures in environmental impact assessments, violation of tribal rights, and absence of viable resettlement plans.
- Result: In 1993, the Government of India formally requested the World Bank to cancel the remaining million loan disbursement, marking an unprecedented victory for grassroots environmental resistance against international development finance.
4. Supreme Court Jurisprudence:
In Narmada Bachao Andolan v. Union of India (2000), the Supreme Court allowed dam construction to proceed incrementally up to 138.68 meters, but established the binding legal doctrine that dam elevation can only be raised pari-passu (step-by-step in parallel) with prior, certified, complete rehabilitation of every affected family.

>>**Solution to Question 14:** the environmental ethos of the Bishnoi Community, the 29 Principles of Guru Jambheshwar, and the 1730 Khejarli martyrdom.

slaughter.
- Royal Decree: The Maharaja issued a royal copper-plate edict strictly banning the felling of green trees and the hunting of any wildlife across all Bishnoi territories in perpetuity, establishing the sacred sanctuary that protects wild Blackbuck, Chinkara, and Khejri forests to this day.

>>Solution to Question 15: Compare the philosophical paradigms of Anthropocentrism, Biocentrism, and Ecocentrism, evaluating Arne Naess' Deep Ecology frame

* 6. Policies affecting basic economic, technological, and ideological structures must change.
* 7. Ideological change is mainly that of appreciating life quality rather than adhering to an increasingly higher standard of living.
* 8. Those who subscribe to the foregoing points have an obligation to participate in the attempt to implement the necessary changes.

The oral Reference Dose (RfD, expressed in mg/kg-day) is an estimate of a daily oral exposure to the human population (including sensitive subgroups such as infants and the elderly) that is likely to be without an appreciable risk of deleterious non-carcinogenic health effects during a lifetime.

2. Determination of Critical Toxicological Endpoints:

Derived from chronic or sub-chronic in-vivo animal bioassays evaluating the most sensitive critical toxic effect:

- NOAEL (No-Observed-Adverse-Effect Level):

The highest experimental dosage administered to test animals at which there is no statistically or biologically significant increase in the frequency or severity of any adverse toxic effect compared to the control group.

- LOAEL (Lowest-Observed-Adverse-Effect Level):

The lowest experimental dosage tested that produces a statistically significant adverse biological effect. Used when a clean NOAEL is not identified in the empirical dataset.

3. Mathematical Derivation of RfD via Uncertainty Factors:

RfD = NOAEL / (UF_total * MF)

Where UF_total is the product of specific 10-fold Uncertainty (Safety) Factors (UF_1 to UF_5), and MF is a Modifying Factor (0.1 to 10):

- UF_A (Interspecies Extrapolation, 10x): Accounts for the uncertainty of extrapolating toxicological responses from experimental animals (rats, mice) to humans, compensating for metabolic and physiological differences.
- UF_H (Intraspecies Human Variability, 10x): Accounts for the biological variation in sensitivity among human beings, protecting genetically susceptible individuals, pregnant women, and infants.
- UF_S (Sub-chronic to Chronic Extrapolation, 10x): Applied when deriving a lifetime chronic RfD from a sub-chronic (e.g., 90-day) animal feeding study.
- UF_L (LOAEL to NOAEL Extrapolation, 10x): Applied when a NOAEL was not observed in the study and a LOAEL must be utilized as the baseline point of departure.
- UF_D (Database Completeness, 3x to 10x): Applied when the underlying toxicological database lacks key bioassays (e.g., missing multi-generational reproductive toxicity studies).

4. Example Calculation:

If a pesticide demonstrates a chronic rat NOAEL = 10.0 mg/kg-day, applying standard 100-fold uncertainty (UF_A * UF_H = 10 * 10 = 100):

RfD = 10.0 / 100 = 0.10 mg/kg-day.

>>**Solution to Question 18:** Analyze the mathematical metrics used in environmental health epidemiology: Relative Risk (RR), Odds Ratio (OR), and Disability-Adju

* RR < 1.0: Exposure is protective.
2. Odds Ratio (OR):
- Definition: Used in retrospective Case-Control Studies where incidence rates cannot be calculated directly because the total exposed population denominator is unknown.
- Formulation:
* Odds of Exposure among Cases (Diseased): Odds_cases = a / c.
* Odds of Exposure among Controls (Healthy): Odds_controls = b / d.
* Odds Ratio: OR = (a / c) / (b / d) = (a * d) / (b * c) (The Cross-Product Ratio).
- Rare Disease Assumption: When a disease is rare in the general population (< 5%), the Odds Ratio mathematically approximates the Relative Risk: OR ~ RR.
3. Disability-Adjusted Life Years (DALY, Murray and Lopez, 1996):
- Formulated by the WHO and World Bank to quantify the Global Burden of Disease (GBD), measuring the gap between current health status and an ideal health situation where the entire population lives to an advanced age, free of disease and disability:
DALY = YLL + YLD
- 1. Years of Life Lost (YLL): Quantifies premature mortality:
YLL = N * L
Where N is the number of deaths attributable to the environmental factor, and L is the standard life expectancy remaining at the age of death.
- 2. Years Lived with Disability (YLD): Quantifies morbidity:
YLD = I * DW * D
Where I is the number of incident cases, DW is the Disability Weight (ranging from 0.0 for perfect health to 1.0 for death), and D is the average duration of the disability in years.

>>**Solution to Question 18:** the molecular toxicology of Lead (Pb) poisoning, explaining ALAD enzyme inhibition, calcium mimicry, and cognitive neurodevelo

- Pb2+ has an ionic radius (1.19 A) nearly identical to divalent calcium (Ca2+, 1.00 A) and readily substitutes for Ca2+ in biological systems, but binds with higher affinity, disrupting calcium signaling:
- Crosses the blood-brain barrier by masquerading as Ca2+ in calcium-ATPase channels.
- In presynaptic nerve terminals, Pb2+ inappropriately activates Protein Kinase C (PKC) and calmodulin, triggering spontaneous neurotransmitter leakage, while simultaneously blocking voltage-gated calcium channels, inhibiting evoked release of acetylcholine and glutamate.
- In postsynaptic neurons, Pb2+ blocks N-methyl-D-aspartate (NMDA) glutamate receptors, completely destroying Long-Term Potentiation (LTP)-the neuro-cellular basis of memory consolidation and synaptic plasticity.
- Clinical Manifestation in Children: Irreversible loss of IQ points (an average 2-4 IQ point deficit per 10 ug/dL increase in blood lead), impaired executive functioning, shortened attention span, aggressive anti-social behavior, and acute Lead Encephalopathy (seizures, coma, death at BLL > 70-100 ug/dL).

>>Solution to Question 20: Examine the low-dose non-monotonic dynamics of Bisphenol A (BPA) and evaluate evidence for Epigenetic Transgenerational Inherita

- BPA induces permanent alterations in DNA methylation, histone post-translational modifications, and small non-coding RNA (sncRNA) profiles in the male and female germline.
- These aberrant epigenetic marks evade post-fertilization epigenetic reprogramming, transmitting adult-onset prostate pathology, polycystic ovarian syndrome, and reduced fertility across three to four unexposed generations.

>>Solution to Question 21: Analyze the architecture of Environmental Justice, the Environmental Generational System (EGS), Climate Equity, and the Basel Convention

Established as a planned program by the Ministry of Environment, Forest and Climate Change (MoEFCC), Government of India, following recommendations of the Tiwari Committee (1980). ENVIS serves as a nationwide decentralized network for comprehensive collection, storage, retrieval, and electronic dissemination of environmental information to scientists, engineers, policymakers, and the public.

2. The Decentralized Distributed Architecture:

ENVIS operates through a decentralized hub-and-spoke organizational architecture coordinated by the ENVIS Focal Point located within the MoEFCC in New Delhi:

- 1. ENVIS Hubs (State Government Nodes):

* Established within State Departments of Environment or State Pollution Control Boards across all Indian states and Union Territories.
* Mandate: Collecting and hosting statewide environmental databases, state-of-environment reports (SoER), status of river catchments, and local municipal solid waste management data.

- 2. ENVIS Resource Partners (Thematic Nodes):

* Established within premier specialized national research institutions, university departments, and environmental centers of excellence:
* Examples: Wildlife Institute of India (Faunal Biodiversity); Botanical Survey of India (Floral Diversity); NEERI (Environmental Engineering & Water Quality); TERI (Renewable Energy); IIT Bombay (Environmental Biotechnology); Forest Research Institute (Forestry Management).
* Mandate: Building specialized national databases on their specific thematic discipline, publishing bibliographic monographs, and responding to technical user queries.

3. Modern Upgrades: Environmental Information, Awareness, Capacity Building and Livelihood Programme (EIACP):

- Aligned with the national Digital India initiative and Mission LiFE (Lifestyle for Environment).
- Functions as an open-access web portal integrating geospatial maps, real-time National Air Quality Index (NAQI) feeds, coastal zone databases, and green technology patents.
- Administers the Green Skill Development Programme (GSDP), training rural and urban youth in solar panel maintenance, river water quality testing, sewage plant operation, and forest monitoring.

>> Solution to Question 23: Explain the applications of Satellite Remote Sensing in environmental management, detailing multispectral sensors, NDVI, and forest m

sharp absorption cliffs.
3. The Normalized Difference Vegetation Index (NDVI, Rouse et al., 1974):
NDVI = (rho_NIR - rho_Red) / (rho_NIR + rho_Red)
- NDVI ranges strictly between -1.0 and +1.0.
- Dense, Healthy Tropical Forest: NDVI = 0.65 to 0.85 (Massive NIR reflection, near-zero red reflection).
- Sparse Grassland / Shrub: NDVI = 0.20 to 0.40.
- Bare Soil / Desert Sand: NDVI = 0.05 to 0.15.
- Water Bodies & Snow: NDVI = Negative values (-0.1 to -0.6) because red reflectance exceeds NIR.
4. Operational Environmental Applications in India:
- Forest Survey of India (FSI): Utilizes multi-spectral sensors (IRS LISS-III, Sentinel-2, Landsat-8) to publish the biennial India State of Forest Report (ISFR), classifying forest canopy cover into Very Dense Forest (> 70% canopy), Moderately Dense (40-70%), and Open Forest (10-40%).
- Tracking Deforestation and Encroachment: Automated digital change-detection algorithms identify illegal logging and forest fires in real time.
- Disaster Monitoring: Flood inundation mapping during monsoons (using Synthetic Aperture Radar SAR from Sentinel-1, which penetrates cloud cover), tracking Himalayan glacial lake outburst floods (GLOFs), and measuring urban heat island expansion.

>>Solution to Question 24: Analyze the role of Geographic Information Systems (GIS) in environmental decision-making, spatial overlay analysis, and watershed m

- Simulates sediment erosion yields and models peak storm hydrographs, guiding civil engineers in positioning optimal check dams and groundwater percolation tanks.

- Centre for Science and Environment (CSE, New Delhi):
* Founded by Anil Agarwal; publishes the authoritative biennial 'State of India's Environment' reports.
* Conducted pioneering investigative research exposing toxic pesticide residues in commercial bottled soft drinks and leading the public campaign for CNG public transit in Delhi.
- Kalpavriksh Environmental Action Group (Pune):
* Specializes in community-conserved areas, wildlife corridor monitoring, and documentation of indigenous rights under the Forest Rights Act.
- Wildlife Trust of India (WTI) and WWF-India: Active in species recovery, emergency animal rescue, and anti-poaching operations.
3. Landmark Environmental PIL Jurisprudence:
- Ganga Pollution Case (M.C. Mehta v. Union of India, 1988): Supreme Court ordered the immediate shutdown of over 30 defaulting commercial tanneries in Jajmau, Kanpur, discharging untreated chromium and organic effluent into the sacred Ganga River.
- Delhi CNG Conversion Case (1998): Court ordered the entire commercial public bus, auto-rickshaw, and taxi fleet of Delhi (over 100,000 vehicles) to convert from diesel to Compressed Natural Gas (CNG), reducing urban sulfur dioxide and particulate smoke.
- Taj Trapezium Case (1996): Sited an air pollution restriction zone (10,400 km2 around Taj Mahal), ordering coal-based glass foundries and iron works to switch to clean natural gas or relocate.

>>P**Solution to Question 27:** Examine the socio-economic and cultural impacts of Development-Induced Displacement, evaluating the marginalization of Project-Affe

- 7. Loss of Access to Common Property Resources (CPRs): Loss of free village commons providing grazing pastures, firewood, medicinal plants, and water bodies.
- 8. Social Disarticulation (Cultural Fragmentation): The destruction of ancient community kinship networks, mutual-aid safety nets, sacred ancestral shrines, and cultural identity.

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model) to predict steady-state downwind pollutant concentrations from an elevated continuous point source (chimney stack):
- Assumptions: Continuous steady emission rate Q (g/s); uniform horizontal wind velocity u (m/s); complete ground reflection (no surface deposition); steady meteorological stability; pollutant behaves as an inert, non-reactive gas; crosswind and vertical concentration profiles follow normal (Gaussian) bell-shaped distributions.
2. Mathematical Formulation:
The pollutant concentration C(x,y,z) at downwind distance x, crosswind distance y, and vertical receptor height z above ground is given by:
C(x,y,z) = [Q / (2 * pi * u * sigma_y * sigma_z)] * exp[-0.5 * (y / sigma_y)^2] * { exp[-0.5 * ((z - H) / sigma_z)^2] + exp[-0.5 * ((z + H) / sigma_z)^2] }
Where:
- Q: Pollutant mass emission rate (g/s).
- u: Mean wind speed at effective stack release height (m/s).
- H: Effective stack release height (m), equal to physical stack height (h_s) plus buoyant/momentum plume rise (Delta h).
- sigma_y: Horizontal crosswind dispersion coefficient (m).
- sigma_z: Vertical dispersion coefficient (m).
Both sigma_y and sigma_z expand continuously as power functions of downwind distance x.
- The second exponential bracket accounts for the 'virtual source' at -H representing 100% upward reflection of the plume off the ground surface.
- Ground-Level Plume Centerline Concentration (y = 0, z = 0):
C(x,0,0) = [Q / (pi * u * sigma_y * sigma_z)] * exp[-0.5 * (H / sigma_z)^2].
3. Pasquill-Gifford Atmospheric Stability Classes:
Atmospheric turbulence and vertical temperature gradients are categorized into six stability classes (A to F) based on surface wind speed and incoming solar insolation (day) or cloud cover (night):
- Class A: Extremely Unstable (strong solar insolation, light winds, vigorous thermal convection).
- Class B: Moderately Unstable.
- Class C: Slightly Unstable.
- Class D: Neutral (overcast day or night, high winds > 6 m/s; mechanical shear turbulence dominates).
- Class E: Slightly Stable (nighttime, moderate radiative cooling).
- Class F: Moderately Stable (clear nighttime, light winds < 2 m/s, strong radiation temperature inversion, near-zero vertical mixing, trapping chimney smoke near ground in dangerous 'Fumigation' plumes).

>> **Solution to Question 38:** Evaluate the engineering principles of Circular Sustainable Cities, evaluating the Kalundborg Industrial Symbiosis paradigm and Urban Me

2. The Kalundborg Industrial Symbiosis Paradigm (Denmark):
The world's preeminent, pioneer benchmark of industrial ecology, established organically over five decades among co-located independent industrial enterprises:
- Primary Industrial Partners:
* Asnaes Power Station (1,500 MW coal/biomass thermoelectric plant).
* Equinor Oil Refinery (large petroleum refinery).
* Novo Nordisk (major global pharmaceutical insulin and industrial enzyme manufacturer).
* Gyproc (commercial gypsum wallboard manufacturer).
* Municipality of Kalundborg (district residential heating network).
- Cyclic Symbiotic Material and Energy Cascades:
* 1. Waste Steam Cascading: Low-pressure waste exhaust steam from the power station is ducted directly to the oil refinery and Novo Nordisk fermentation plant, eliminating separate fossil fuel boilers. Surplus residual steam supplies the Kalundborg district heating network, eliminating 3,500 domestic oil-fired home furnaces.
* 2. Synthetic Gypsum Exchange: Flue gas desulfurization (FGD) scrubbers at the power plant react SO_2 with limestone to precipitate high-purity synthetic gypsum ($CaSO_4.2H_2O$). This chemical by-product is piped directly to Gyproc, completely replacing the need to import mined natural gypsum from Spain!
* 3. Wastewater Cascades: Treated cooling water and wastewater from the oil refinery is piped to the power plant for boiler feed, slashing local freshwater abstraction.
* 4. Biomass Sludge Fertilizer (Novogro): Fermentation broth residues containing deactivated yeast and fungal mycelia from Novo Nordisk are pasteurized and distributed as high-nitrogen organic bio-fertilizer to over 800 surrounding regional agricultural farms, replacing synthetic Haber-Bosch chemical fertilizers.
3. Lessons for Biotechnology and Future Sustainable Cities:
Demonstrates that industrial biotechnology, manufacturing, and municipal urban communities can co-locate to form closed-loop industrial ecosystems, cutting carbon emissions by hundreds of thousands of tons while generating millions of dollars in shared economic profits.
